ModelIt!

Student Activity Pack

Slime Is Serious Science

How Molecules Determine the Properties of Everything You Touch

Grade 6 | NGSS MS-PS1-1, MS-PS1-2 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Is slime a solid, liquid, or gas? Explain your reasoning.

2 What do you think molecules look like inside slime compared to inside water?

3 Why do you think some liquids (like honey) are thick while others (like water) are thin?

4 Draw what you think is happening at the molecular level when you stretch slime.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Molecule Size | Bond Strength | Temperature | Concentration

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Liquid Slime

Low concentration, weak bonds

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Perfect Slime

Moderate concentration and bond strength

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Solid Slime

Maximum concentration and bond strength

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Cross-Linking Density — The number of chemical bridges between polymer chains — more cross-links make the substance stiffer

Solvent Type — Water vs. other liquids — different solvents interact differently with polymer chains

Additive Particles — Small particles added to the mixture can change its texture, color, and flow properties

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does slime act like both a liquid AND a solid at the same time? Use evidence from your simulation to support your answers.

1 Why does slime stretch slowly but break when you pull it quickly?

2 How is making slime a chemical change, not just a physical change?

3 If you could design a molecule for the perfect bouncy ball, what would it look like?

4 Why does warming up slime make it runnier? What's happening to the molecules?

5 How does understanding molecular structure help engineers design new materials?

8 STEM Challenge: Design the Perfect Slime Formula

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YOUR ENGINEERING MISSION

Engineer a slime with specific target properties (stretch distance, bounce height, flow rate) by adjusting your molecular recipe based on model evidence.

THE SCENARIO

A toy company needs a new slime product. They want three versions: 'Super Stretch' (maximum elongation), 'Bouncy Blob' (maximum bounce), and 'Slow Flow' (slowest pour). Your chemistry team must design formulas for all three using your molecular model.

Engineering Design Thinking

1. WHAT RATIO OF GLUE TO ACTIVATOR WILL YOU USE FOR EACH SLIME TYPE?

2. HOW WILL YOU MEASURE STRETCH DISTANCE, BOUNCE HEIGHT, AND FLOW RATE CONSISTENTLY?

3. WHAT HAPPENS IF YOU CHANGE THE TEMPERATURE OF YOUR SLIME? DOES IT HELP OR HURT?

4. HOW WILL YOU RECORD AND COMPARE YOUR RESULTS FOR THE THREE FORMULAS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Materials Scientists and Polymer Chemists design new materials with specific properties for products like sneakers, phone cases, medical devices, and more. They earn $70,000–$130,000/year.